An operator assistant for old cooling-water pumps

Team detector brief for the jury. RETROFIT AI challenge, Since AI Hackathon 2026, Turku. Challenge owner Business Turku, technical partner Elomatic.

84
our composite score, all 9 labelled pumps

63
the organizers' baseline on the same pumps

0
false alerts and excess repeats

27 / 35
faults named with the right cause (baseline: 11 / 35)

In one paragraph. Every pump comes with a factory data sheet saying how much pressure and power it should give at a given speed. Our program compares what the pump actually does with that sheet, every minute. When the two drift apart in a pattern an engineer would recognise, it raises one alert that names the likely cause, says how sure it is, and says what to go and check. Before blaming the pump, it checks whether one sensor is simply lying, by comparing it with the others. It never controls the machine.

The problem

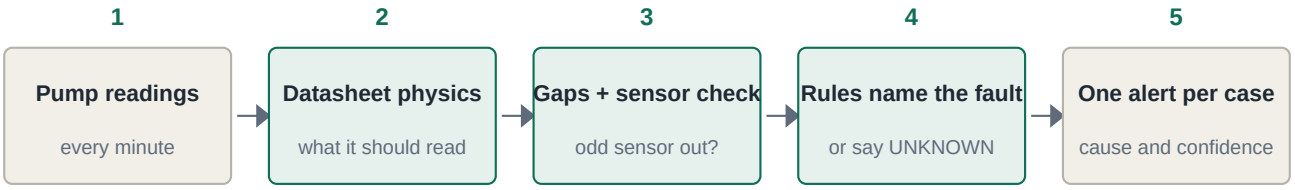
Pumps installed 10 to 30 years ago work, but cannot say when they are starting to fail; an operator finds out when a fixed alarm limit is crossed, usually too late to plan the repair. The challenge asks for an assistant that reads the data an old pump already produces, notices developing problems early, explains the evidence, says how certain it is, and recommends what a human should check. The data: twelve synthetic pumps, seven days each, one reading per second, sensor levels from four signals to fourteen. Nine pumps come with labelled faults; three are hidden and scored by the organizers.

Problem	What an operator would see
Strainer clogging	Inlet pressure and flow fall; a variable-speed drive speeds up to compensate
Bearing wear	High-frequency vibration rises first, later overall vibration and heat
Cavitation	Inlet pressure near the boiling point for that water; noisy vibration, lost pressure
Discharge restriction	Outlet pressure up, flow and current down
Misalignment, imbalance	Vibration up, especially with speed
Impeller wear, air in the water	Pressure lower than the pump should give at that flow
Sensor problems	One signal changes while related signals do not: drift, frozen value, offset step, noise, data gaps

Telling a machine fault from a misbehaving sensor is a large part of the challenge, and where the baseline is weakest: it names the right cause for 27 % of the training faults. All numbers here come from the organizers' scoring tool on their published labels. Code: github.com/timursdobrovolskis-cyber/since-ai-hackathon, branch *detector*.

1. The solution in plain words

There is no trained model inside. The detector is the pump's data sheet plus a dozen rules written the way an engineer would say them. That is deliberate: every alert can be explained in one sentence, it cannot peek into the future, and it transfers to any pump that has a data sheet.



Green boxes are the part we built. No model is trained: it is the pump's data sheet plus written rules.

Three examples of what it sees

Situation	What the signals do	What the alert says
Clogged filter	Inlet pressure drops below what it should be at this flow, while the pump itself still behaves exactly like its data sheet	"Suction pressure 9 kPa below normal for 294 m^3/h; head still on the pump curve. Inspect and clean the suction strainer at the next opportunity."
Failing bearing	The high-pitched vibration climbs first, long before the pump shakes or heats up	"Vibration acceleration 0.3 g (z=+5.2), crest factor up: high-frequency energy first, typical of a rolling-bearing defect. Plan a bearing replacement."
Lying flow meter	The meter says one thing, but the pressures and the motor power agree on another	"Flow meter 61 m^3/h off the flow implied by power and by head, which agree with each other; it stepped in one move 15 min ago. Compare with a local gauge."

How the work was done

Four phases, each scored on the same training and validation pumps with the organizers' tool before moving on. First reproduce the baseline exactly. Second, keep its alert times and replace only the diagnosis with physics. Third, replace its alert timing with our own. Fourth, calibrate the confidence. The score went 63, 76, 84 on all nine pumps, and the baseline's false alerts went to zero.

2. The physics behind it

The organizers allow and encourage the use of the pump curve in the asset list. We use it as the reference for everything.

Expected readings from the data sheet

A centrifugal pump's head (pressure rise) at a given flow and speed follows its curve: shut-off head, best-efficiency point and the affinity laws (flow scales with speed, head with speed squared, power with speed cubed). From the data sheet alone we compute, every minute, the head, power and current the pump should show at its current speed and flow. Motor and drive losses follow standard catalogue relations.

Flow three ways

Flow is estimated from the flow meter (where there is one), from the measured head through the curve, and from the motor power or current. Three independent estimates of one quantity are what make the sensor check possible: if two agree and one does not, the odd one is the instrument at fault.

Each pressure against its own system curve

The inlet pressure of a healthy pump falls with flow squared along a fixed line, and so does the outlet pressure rise. We fit both lines during the pump's first day and watch the departures. A clogged strainer pulls the inlet pressure below its line while the pump stays on its own curve. A restriction pushes the outlet pressure above its line at the same flow. Where the plant measures the suction level (cooling-tower and process pumps), the level enters the inlet line at 9.8 kPa per metre, so a basin running low is not mistaken for a clog.

Cavitation margin

From inlet pressure, liquid temperature (vapour pressure) and flow we compute the net positive suction head available and compare it with the pump's requirement from the data sheet. A margin below 1.3 with rising acceleration or lost head is cavitation. This is the same definition the simulator uses to label episodes, and it catches every cavitation episode in the training data before the strainer was cleaned; the baseline caught none.

Fault signatures used by the rules

Named cause	Signature in the residuals
Suction strainer clogging	Inlet pressure below its system line; head still on the pump curve; on a drive, speed and power up
Cavitation	NPSH margin below 1.3 with high acceleration or head loss
Discharge restriction	Outlet pressure above its system line while power, current or flow are down
Air entrainment / impeller wear	Head below the curve with a fine NPSH margin: sudden (within 8 h) is air, slow is wear
Bearing wear	Acceleration or crest factor up; velocity and bearing temperature follow later
Misalignment / imbalance	Velocity up without acceleration: misalignment if it began within 36 h of a maintenance restart, else imbalance

3. Telling a bad sensor from a bad pump

A sensor fault changes one reading. A machine fault changes several readings together, in the way physics dictates. The watchdog uses that difference, and it runs before the machine rules, because a bad pressure transmitter would corrupt every head calculation downstream.

Instrument fault	How it is recognised	Example from the labelled data
Frozen reading	A run of identical values longer than twice the longest run seen while learning, while the pump runs and load varies	PUMP-01 bearing temperature constant for hours; found 36 min after it froze
Noise and spikes	The reading jumps within each minute three times more than normal while related signals are calm	PUMP-08 discharge pressure spikes after a cable fault; found in 6 min
Calibration step	One instrument moves off its physics prediction in a single jump while the others agree	PUMP-07 flow meter after maintenance; found in 12 min, named correctly
Drift	The same, but creeping up over hours	PUMP-02 suction transmitter drifting over four days; named while a bearing was also failing
Communication loss	Rows stop arriving from the gateway	Both outages in the labels found within 15 min

Why this matters for the operator

A false "clogged strainer" sends a crew member to open a strainer basket at night. A false "sensor fault" sends a technician with a reference gauge. Getting the class right is a 10 % line item in the organizers' score, but in a real engine room it is the difference between useful and ignored. Our class accuracy is 81 % on training pumps and 89 % on validation; the baseline's is 54 % and 89 %.

What the baseline got wrong here

The baseline treats any single channel that deviates as a "calibration shift". It said that eight times on the training pumps and was right once. It also called rising vibration a sensor fault twice. Vibration rising on its own is a machine telling you something; we never treat it as an instrument problem.

Guards that keep the watchdog honest

Temperatures are ignored for two hours after a restart, because they are still warming up. Load-side checks (flow versus power) are skipped under heavy vibration, because a failing bearing really does draw extra power. Liquid temperature alone is never called a sensor fault, because on a ship it follows the sea.

4. Alerts and confidence you can trust

The scorer penalises false alerts and constant re-alerting, and so does any crew. The alert policy is therefore as important as the diagnosis.

Alert policy

Rule	Why
A deviation must hold for 15 minutes of steady running	Start-ups, speed changes and load steps look abnormal for a few minutes and are not faults
Nothing alerts in the first 24 hours	That is the learning period; alerting inside it would use data from the future. The baseline does this and would fail the organizers' truncation test
One alert per case, a reminder after 12 hours	A case is one machine fault type, or one instrument whatever it does
Episodes re-arm when their evidence clears	Cavitation, air and communication loss come and go; each episode is a separate labelled event
UNKNOWN only after three hours, never on top of an open case	Honest when unsure, without burying a named alert under vague ones
Alerts at least 15 minutes apart, no bursts	Listing several guesses at once earns nothing and confuses the operator

Confidence that means something

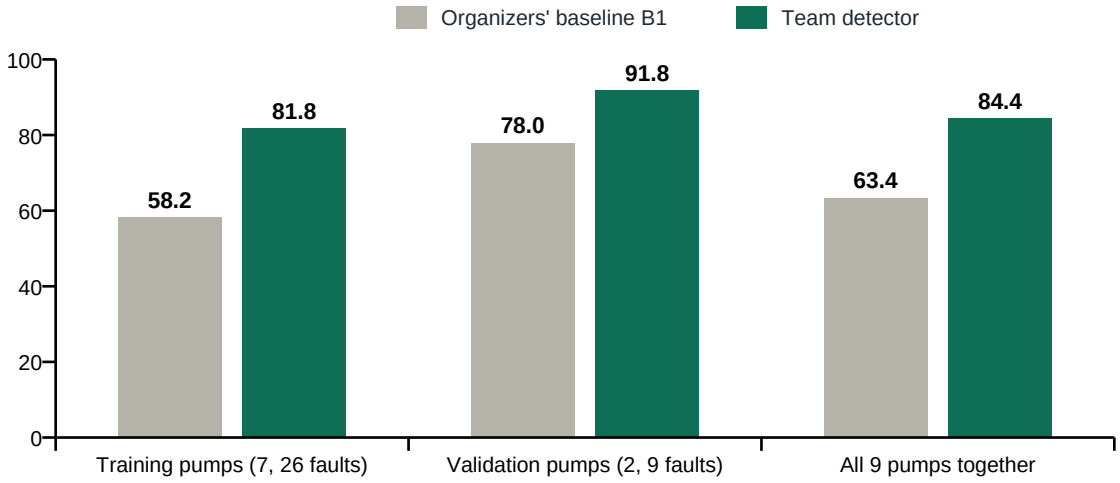
The confidence on each alert is not a tuning knob. For every fault name it is the share of alerts with that name on the training pumps that fell inside a labelled event of that type, with light smoothing. It is fitted on the seven training pumps only and then checked on the two validation pumps the detector had never seen. There, the observed hit rate was at or above the stated confidence for every fault type, so the numbers are conservative rather than flattering.

Named cause	Stated confidence	Observed on validation
Bearing wear	0.95	1.00 (5 alerts)
Strainer clogging	0.90	1.00 (3)
Discharge restriction	0.86	1.00 (4)
Sensor stuck	0.86	1.00 (2)
Misalignment	0.43	1.00 (5)
Sensor drift	0.53	no validation alerts
UNKNOWN	0.50, flagged as not calibrated	none raised

Misalignment is a good example of honesty: on the training pumps two of five misalignment calls were wrong, so the detector says 0.43 even though it was right every time on validation. The app shows these as High, Medium or Low, and the alert file marks the one value that is not calibrated.

5. Measured results

All numbers are from the organizers' scorer on their published labels: seven training pumps with 26 scored faults and two validation pumps with 9. Settings were chosen on the training pumps only; validation was used to check, never to fit. Validation has few events, so the organizers ask teams to read both splits together.



Metric	B1 train	Team train	B1 valid.	Team valid.	B1 all	Team all
Composite score (0-100)	58.2	**81.8**	78.0	**91.8**	63.4	**84.4**
Faults detected	18 / 26	**21 / 26**	8 / 9	8 / 9	26 / 35	**29 / 35**
Right cause named	7 / 26	**19 / 26**	4 / 9	**8 / 9**	11 / 35	**27 / 35**
Machine vs sensor right	0.54	**0.81**	0.89	0.89	0.63	**0.83**
Timeliness (1 = before the limit)	0.58	**0.71**	0.86	**0.89**	0.65	**0.75**
False alert episodes	3	**0**	1	**0**	4	**0**
Excess repeat episodes	3	**0**	0	0	3	**0**
Precision of alerts	0.75	**1.00**	0.89	**1.00**	0.79	**1.00**

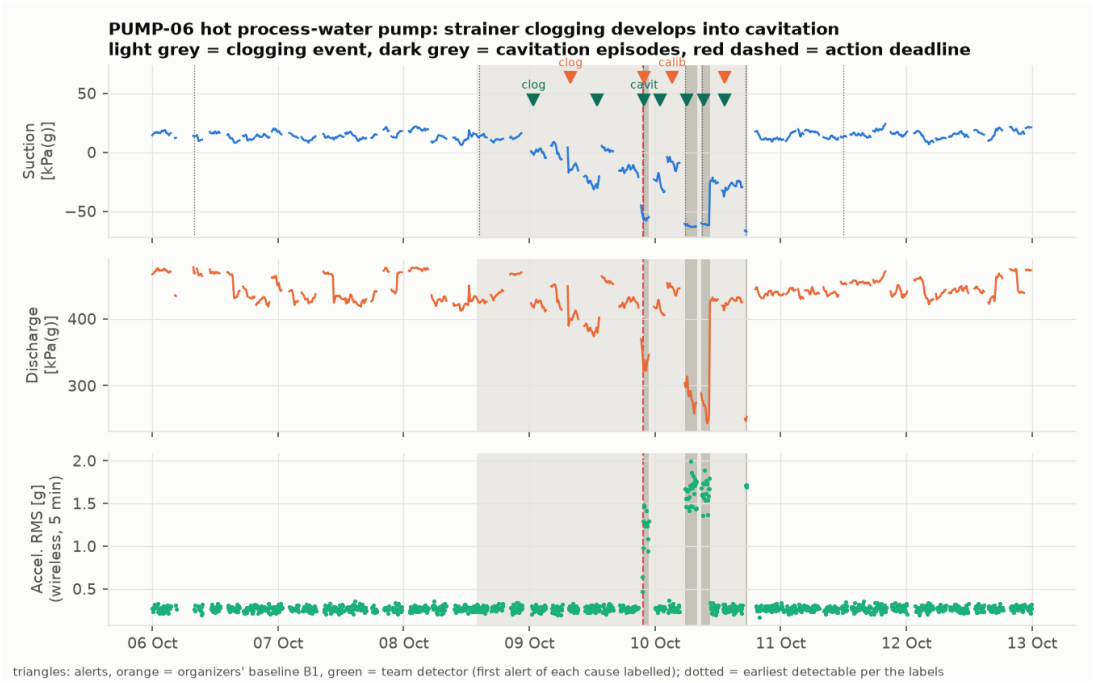
Early warning

Where a conventional alarm limit exists in the labels, the detector warns before it in every case it catches: a cooling-tower pump's bearing 86 hours before the limit, a ship's strainer 19 hours before, a restriction 14 minutes before. On validation the median lead is 9.8 hours. The baseline found the training ship's bearing wear after 118 hours; we find it at 36.

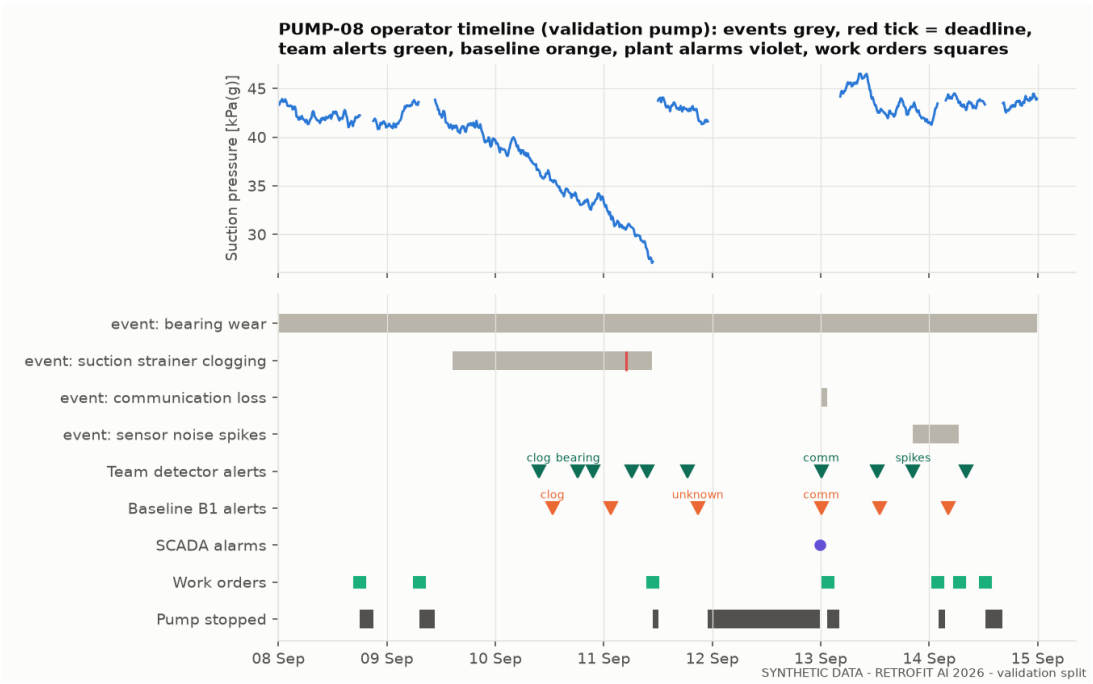
The plant's existing alarms, for reference, score 27.6 on training and 23.1 on validation.

6. Both detectors on the organizers' own figures

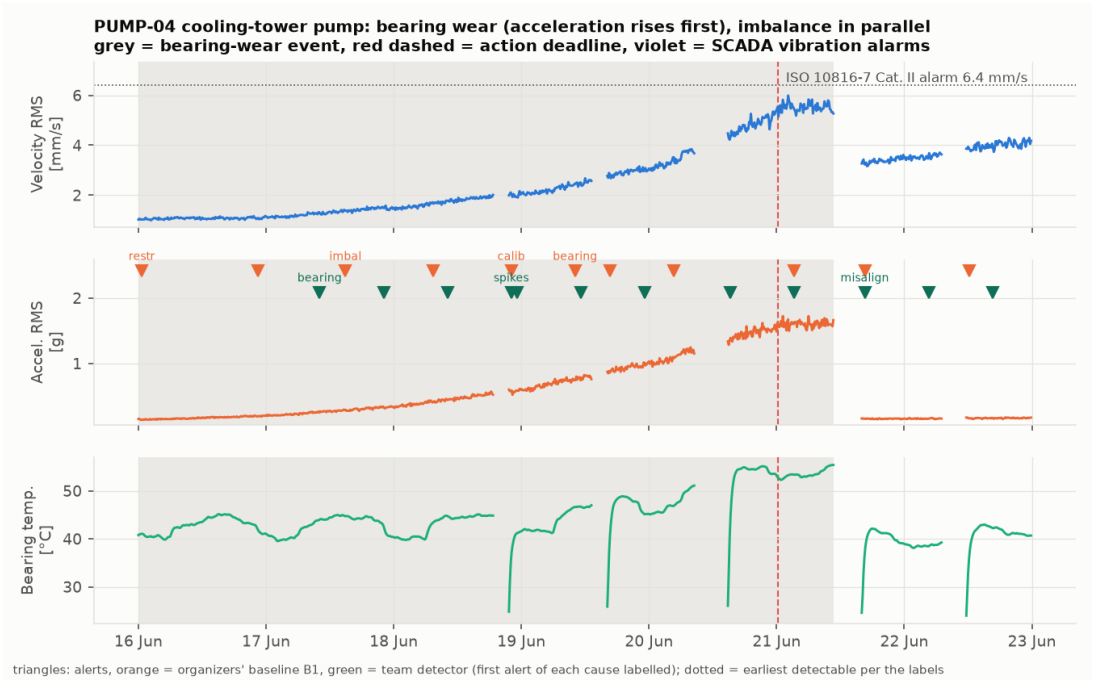
The organizers published figures of four labelled incidents. We redrew them with their code and added every alert: orange triangles are the baseline B1, green triangles are ours, each cause labelled at its first alert. Grey bands are the labelled events, the red dashed line the conventional action deadline.



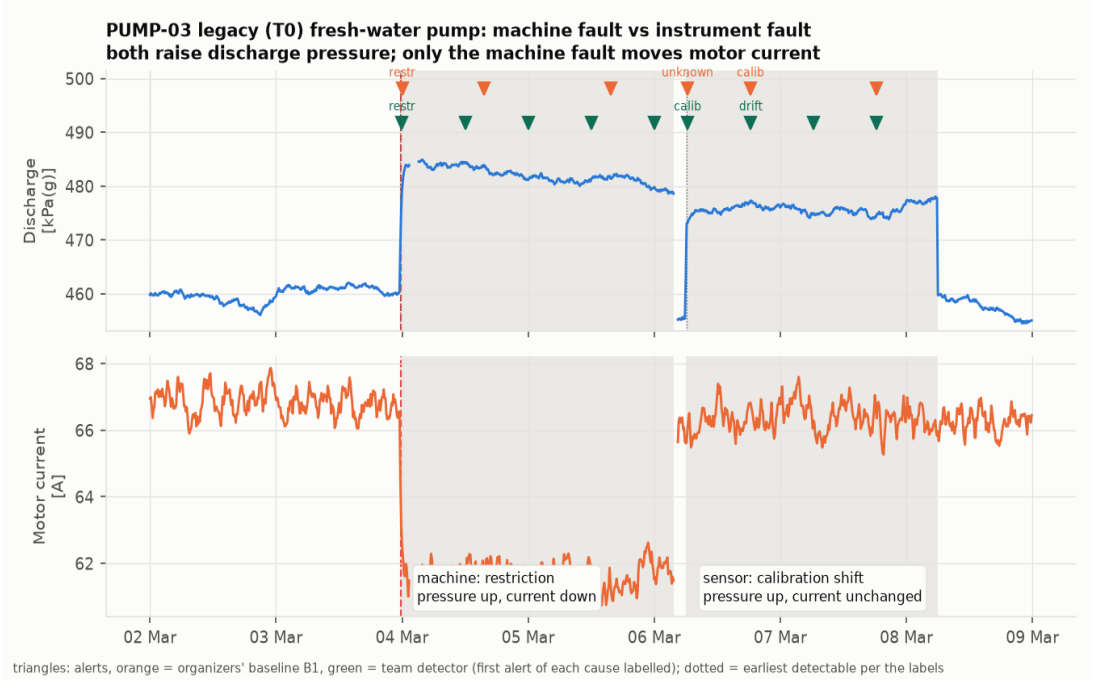
PUMP-06, strainer clogging into cavitation (training). Both name the clog, ours seven hours earlier (01:00 vs 07:54 on 9 Oct), 21 h before the action deadline. The baseline keeps repeating "clog" through the cavitation; ours raises a cavitation alert within 20 min of each of the first three episodes, from the NPSH margin and the acceleration. The 17-minute episode just before the cleaning is missed by both.



PUMP-08, the validation ship pump. The strainer alert comes at 09:36 on 10 Sep (baseline 12:38), 19 h before the limit, with no plant alarm at all. Our second finding is the bearing, which the baseline only ever calls "unknown". Both catch the network outage; ours names the transmitter spikes on 13 Sep, the baseline says "unknown". Reminders are 12 h apart, so none counts as excess.



PUMP-04, bearing wear under a growing imbalance (training). The baseline's first two alerts, inside its own learning window, call it a restriction and an imbalance; it names the bearing only on 19 Jun. Ours stays silent for the learning day, then names the bearing on 17 Jun from the acceleration alone, 86 h before the ISO limit and four days before the replacement. Honest note: after the replacement the remaining vibration is the imbalance, which we call misalignment because the pump had just been reassembled.



PUMP-03, machine fault vs instrument fault on a legacy pump (training). Only pressure and current exist. Both catch the restriction at once (pressure up, current down). When the transmitter offset appears after the repair, the baseline says "unknown", then "calibration shift"; ours says "calibration shift" at once, because the pressure moved in one jump while the current did not. Neither finds the slow impeller wear present from day one.

7. The challenge criteria, one by one

The organizers' composite score (left) is the measured-performance evidence inside the jury's technical criterion. The jury criteria (right) cover the whole project; the detector contributes to each as described.

Composite score weights

Component	Weight	Ours (train / valid.)	Baseline
Event F1: catch faults, no false alerts	35 %	**0.89 / 0.94**	0.72 / 0.89
Fault-type recall: name the cause	25 %	**0.73 / 0.89**	0.27 / 0.44
Timeliness: before the conventional limit	20 %	**0.71 / 0.89**	0.58 / 0.86
Class recall: machine vs sensor	10 %	**0.81 / 0.89**	0.54 / 0.89
Alert burden: few false or repeated alerts	10 %	**1.00 / 1.00**	0.94 / 0.97

Jury criteria

Criterion	Weight	How the detector meets it
Business usefulness	25 %	Every alert names one cause, a confidence an operator can act on, and a concrete check. One alert per problem with a 12-hour reminder: no alarm flood. Early warning turns an unplanned stop into a planned one.
Technical validity and measured performance	25 %	Beats the baseline on both splits (84 vs 63 on all labelled pumps) with the organizers' own scorer. Fitted on train, checked on validation, never touched the test pumps. Every number is reproducible in ten documented steps.
Reliability and safety	20 %	Advice only. Causal by construction and verified by rerunning on data cut at 27 random points. Says UNKNOWN when unsure. Confidence calibrated and conservative on unseen pumps. Sensor faults separated from machine faults so a lying transmitter does not trigger a repair.
Low-cost retrofit feasibility and transfer	20 %	Runs on the legacy tier (four signals from the existing starter panel) and on the cheap wireless vibration tier with five-minute readings. Needs only the pump's data sheet, so it transfers to any catalogue pump. Rules, not a model: no retraining per site.
Working demo and clarity	10 %	The team's app runs on these alerts, with the evidence sentence, the confidence band, a review view against the plant alarms, and a job card per alert.

Hard rules: alerts use only data up to their own time; no hand-labelling or tuning on PUMP-10 to 12; submission in the official format with one cause per alert; no control of any device; no secrets in the code.

8. Where we came from, and why we changed course

The team's earlier experiments, documented in a separate engineering README, tried to *learn* the pump: gradient-boosted (CatBoost) regressors to predict healthy behaviour and flag residuals, then twelve CatBoost classifiers, one per fault, trained on the labelled minutes. All scored with the official tool on the same validation pumps.

Approach (validation pumps)	Composite	Detected	Right cause
Organizers' baseline B1	78.0	8 / 9	4 / 9
B1 + CatBoost classifiers	65.1	7 / 9	3 / 9
B1 + guarded classifiers	72.4	7 / 9	4 / 9
Learned healthy model, residual detector	52.7	6 / 9	0 / 9
Residual detector + CatBoost classifiers	51.1	5 / 9	2 / 9
Physics detector (this brief)	**91.8**	**8 / 9**	**8 / 9**

Why learning did not work here, in the README's own analysis

Only 5,639 minutes on four pumps qualified as healthy training data, none on a legacy pump, so the learned "healthy" model was extrapolating most of the time. The faults overlap (24 pairs on seven pumps), so one pattern carries several labels. Thirteen diagnoses from 26 incidents is too few independent examples for a classifier, and minute-level predictions are not the event-level alerts the scorer wants.

Why the physics route answers each of those

The data sheet *is* the healthy model, for every pump including the legacy ones, with no training minutes. Rules work at the event level, one alert per case. Overlapping faults are handled because each rule reads its own physical signature (a clog in the suction line, a bearing in the acceleration), so two faults are named one after the other. And the rules are the explanation.

The README asked for eight audits before trusting the 91.8

Audit item	Answer
1. Code and dependencies	detector/ on branch detector; pandas, numpy, scipy from the organizers' requirements file
2. Original files unchanged	Our rerun of B1 matches the organizers' shipped validation file row for row
3. Official scorer unaltered	*python -m retrofit_sim score* from the organizers' package gives 91.8 and 81.8
4. Validation used for tuning?	Thresholds swept on train only. Rule choices were checked against validation, as the rules allow for model selection, so validation is not fully independent; train (81.8) and the hidden pumps are
5. Future data	None: silent learning window, backward-looking features, verified by 27 random truncations
6. Labels as inputs	Never. Labels only score the result and calibrate the confidence numbers on train
7. Reproducible	Ten documented steps, deterministic, byte-identical files on rerun
8. Holds on hidden pumps?	Unknown until scored; nothing tuned on them, only alert counts looked at

9. Limitations, honestly

The remaining gap to a perfect score is 18 points on training and 8 on validation. We know where each point went, and in most cases we chose to leave it there.

Where the points go	Train	Why, and why we left it
Five faults never caught	about 10	Minor, slow faults on the pumps with the fewest sensors: pre-existing impeller wear on two legacy pumps that only report pressure and current, a bearing-temperature drift of a few degrees, a current transducer creeping a few tenths of an amp, and a 17-minute cavitation flare. Catching them means lowering the trigger, which adds false alerts on the training pumps, and the scorer penalises false alerts twice. A quieter detector is the better operator tool.
Two faults named wrong	about 3	Both on one pump where a bearing was failing badly at the same time; the imbalance and the blocked discharge were hidden under that vibration. A detector that names one cause per alert will pick the loud one.
Later on faults present from day one	about 4	We stay silent for the first 24 hours while learning what normal is. The baseline alerts inside that window, using data from the future, which the organizers forbid.
One validation fault the baseline caught by luck	1 (valid.)	B1 matched a current drift with a false winding-temperature alert raised inside its own learning period. We removed that behaviour on purpose.

Other limits worth stating

Imbalance and misalignment are separated by maintenance context (a restart within 36 hours), not yet by their different speed dependence. The suction level is used only where the plant measures it; heavy weather on a ship, which also disturbs suction, is not modelled. Confidence is calibrated on 63 training alerts, so rare causes carry a cautious 0.67 from a single hit. And everything is tuned on one synthetic simulator: these results show the method works on a physically coherent benchmark, not field performance.

Safety posture

The detector only advises. It cannot send a command. It prefers silence to a weak guess, says UNKNOWN when the pattern is unclear, and states a confidence that was conservative on pumps it had never seen. When a transmitter is found to be bad, hydraulic diagnoses are suppressed for twelve hours so a lying gauge cannot send a crew to clean a strainer that is fine.

10. Reproduce every number in ten steps

The repository contains a verification sheet (detector/VERIFY.md) with each command and the output to expect. The whole run takes about ten minutes on a laptop with the organizers' data pack next to the repository.

Step	What it shows	Expected
1. Create the Python environment	Standard packages from the organizers' requirements file	installs without error
2. detector/baseline_b1.py	Our rerun of the baseline is identical to the organizers' shipped file	matches shipped file: True
3. detector/run.py	Runs the detector on all 12 pumps, writes the alert files	113 alerts, 85 labelled, 28 test
4. detector/score_all.py	Baseline and ours, train and validation, one table	58.2 / 78.0 vs 81.8 / 91.8
5. python -m retrofit_sim score	The organizers' scorer, none of our code involved	COMPOSITE = 91.8 / 100
6. detector/causality_check.py	Rerun on data cut at 27 random points; alerts before each cut must match	causality check: PASSED
7. predictions_test.csv	The submission file for PUMP-10 to 12, official format	28 rows, one cause each
8. Open team_alerts.csv	Read alerts as an operator would: evidence and action columns	one sentence per alert
9. Build the app data, npm run dev	The app running on our alerts, review view against plant alarms	composite 81.8 / 91.8 shown
10. npm test, npm run build	The app's own checks with the rebuilt data	31 tests pass, build clean

What is in the repository

detector/physics.py (data-sheet pump model), features.py (residuals and watchdog), diagnose.py (the rules), detect.py (alert policy), calibrate.py (confidence), score_all.py and causality_check.py (verification), README.md (method and limitations), VERIFY.md (this sheet). Output files: team_alerts.csv and .json, predictions_test.csv, detector/results/final_scores.md with confusion tables.

Closing

The detector is not smarter than an engineer. It is an engineer who reads the data sheet, watches every pump every minute, never gets tired, and says plainly how sure it is. On the organizers' benchmark that is worth 84 points against the baseline's 63, with no false alarms. On a real ship it would be worth a planned repair instead of a night-time stop.